

Multivariate Financial Dependence Analysis of Asian Markets Using Vine Copulas

Abhay K Singh¹, David E Allen², Robert J Powell¹ and Krishna Reddy³

¹*School of Business, Edith Cowan University, Perth, Western Australia*

²*Centre for Applied Financial Studies, Adelaide, University of South Australia, School of Mathematics and Statistics, Sydney, University of Sydney*

³*Waikato Management School, The University of Waikato New Zealand*

Abstract

Modelling financial dependence is one of the major areas of research in quantitative finance and econometrics. Efficient quantification of dependence is desirable for portfolio theory, hedging, valuation of assets and risk management in general. Quantification of co-movement or multivariate dependence between various international stock markets results in better diversification which benefits investors as it helps in profitable risk distribution. Due to their phenomenal growth in recent years, Asian stock markets have become important for global investors. The study of dependence among Asian stock markets has gained importance as these markets promise diversification benefits. In this paper we study the multivariate dependence structure of eleven Asian financial markets, including Thailand, Malaysia, Indonesia, Singapore, Philippines, Korea, Japan, China, Hong Kong, Taiwan, and India using sophisticated and recently developed model, Regular Vine (R-Vine) Copula. R-Vine Copulas constitute a flexible kind of multivariate dependence models which offer more flexibility than standard multivariate copulas. We also introduce the international markets of Australia, UK and US to the dependence analysis in order to study the influence of these international stock markets on the Asian markets.

1 Introduction

Modelling and understanding the dependence between financial asset returns has a key role in studying asset allocation and investment strategies. Financial risk management across nations is strongly affected by the dependence between international financial markets. It is essential for investors to model and quantify the structure and extent of dependence across these international financial markets. The most commonly used dependence measure for a bivariate relationship is the linear correlation coefficient which is based on the assumption of a joint Gaussian distribution. As

seen in various previous studies (Mandelbrot, 1963; Longin & Solnik, 2001; Karmann & Herrera, 2014; Ang & Chen, 2002, among many others) financial data does not usually follow Gaussian or normal distribution, and the inference regarding the dependence based on correlations often leads to misleading results for non-Gaussian data. Embrechts, McNeil, and Straumann (2002) highlight several properties and pitfalls of using correlation as a dependence measure. Furthermore, linear correlation only measures the degree of dependence between financial markets which quantifies the strength of the relation, but it can not model the dependence structure, which describes the way in which the two markets are associated. Copula overcomes this problem by providing an ideal tool in studying both the degree and the structure of the dependence.

The capability of copula to model and estimate multivariate distributions comes from *Sklar's Theorem* (Sklar, 1959), according to which each joint distribution can be decomposed into its marginal distributions and a copula C responsible for the dependence structure. Conversely, a copula produces a multivariate joint distribution combining the marginal distributions and the dependence between the variables. This makes copulas attractive, as the univariate marginal behaviour of random variables can be modelled separately from their dependence. Copulas prove to be very robust in modelling the joint distribution as they do not require the assumption of joint normality. Joe (1997), Nelsen (1999), Alexandar (2008), Cheung (2009) and Patton (2006) give a useful overview of copulas for financial practitioners.

Although there are different type of bivariate copulas available which exhibit flexible and complex dependence patterns, the choice of multivariate copulas is rather limited. Originally proposed by Joe (1996) so called pair copula constructions (PCC) overcome this issue. Since the introduction, PCC has been further explored and extended by Bedford and Cooke (2001, 2002) and Kurowicka and Cooke (2006). Derived from PCC, vines are flexible multivariate dependence models which are formed using bivariate copulas. Aas, Czado, Frigessi, and Bakken (2009) provided key insights into the special cases of canonical vine (C-Vine) and drawable vine (D-vine) to deduce multivariate copulas based on PCC. Although C and D vines are flexible multivariate copula, they have a restrictive dependency structure on the data in high dimensions. Regular vines (R-vines) are another class of hierarchical graphical model classifying PCC which is less restrictive than C- and D-vines. Dissmann (2010) laid out the inference of R-vines using graph theoretic algorithms.

The added flexibility and dimensions in vine copulas has resulted in a growing interest in its use in financial risk modelling and research. Brechmann and Czado (2013) and Allen, Ashraf, McAleer, Powell, and Singh (2013) recently applied R-Vine copulas to study the financial risk and return dependence between the stocks of the Euro Stoxx-50 and the Dow Jones Industrial Average respectively. Brechmann, Czado, and Paterlini (2014), Low, Alcock, Faff, and Brailsford (2013), Schirmacher and Schirmacher (2008), Aas et al. (2009), Fischer, Kock, Schluter, and Weigert (2009), Aas and Berg (2009), Heinen and Valdesogo (2009) are some other illustrative applications of vine copulas.

The aim of this study is to model the dependence structure of eleven Asian financial markets

including five ASEAN (Association of South East Asian Nations) markets (Thailand, Malaysia, Indonesia, Singapore, Philippine), five east Asian markets (Korea, Japan, China, Hong Kong, Taiwan) and India using R-Vine Copula. The paper also studies the influence of some of the major developed international financial markets of Australia, UK and US on these Asian markets. This study examines two different time periods covering the Asian Financial Crisis (AFC) of 1997-1999 and the Global Financial Crisis (GFC) of 2007-2009 to compare the dependence structure between the Asian markets and the influence of the other developed markets during these different financial crisis events, thus contributing to the limited number of studies using vine copulas to model financial risk. The choice of the data time period aligns with the major objectives of the study; to study the dependence structure of developing Asian financial markets along with developed financial markets around the time of financial distress. As previously noted, financial returns do not usually follow a normal distribution and they develop non-normality particularly during the time of financial distress or upturn. Copula provide models to capture dependence between return distributions which are not Gaussian in nature and with the flexibility of vine copulas it is now possible to model these dependence in a high dimension more efficiently than the multivariate Gaussian or Student-t, as well as exchangeable Archimedean copulas.

Since 1990s and thereafter, the growing importance of Asian capital markets due to their phenomenal economic growth has resulted in numerous studies focussing on the dependence between Asian markets and the stock markets in developed countries. A. M. Masih and Masih (1999), Ghosh, Saidi, and Johnson (1999), R. Masih and Masih (2001), Ng (2000), Johnson and Soenen (2002), Park (2010), Chow, Huang, and Niu (2011) and Palamalai, M, and Devakumar (2013) are some previous studies examining the dependence between the financial markets of emerging Asian countries and developed countries. These studies used linear or dynamic correlation, regression, cointegration, GARCH and Vector Error Correction type models to quantify the dependence between these financial markets. To the best of our knowledge there is no previous study using R-vine copula to study the dependence structure of these markets. Sriboonchitta, Liu, Kreinovich, and Nguyen (2014) used C- and D- vine copula approach to study the dependence in the stock markets of Indonesia, Philippine and Thailand, and other than this there is no prior study using vine copula models to study the dependence structure of Asian financial markets.

The paper is divided into five sections: the next section provides an overview of vine copulas, in particular R-vine copulas, section 3 introduces the data and methodology, section 4 presents the results and a brief conclusion follows in section 5.

2 Vine Copulas

As mentioned earlier in the introduction, a vine is a flexible graphical model for describing multivariate copulas built up using a cascade of bivariate copulas, the so-called PCC. Aas et al. (2009), provided the major statistical inference into C-vine and D-vine categories of vine copulas which

belong to the more generalised R-vine copulas. R-vines can be depicted in a graphical theoretic model to determine which pairs are included in a pair-copula decomposition. An R-vine is a special case for which all constraints are two-dimensional or conditional two-dimensional. Regular vines generalize trees, and are themselves specializations of Cantor trees. Combined with copulas, regular vines have proven to be a flexible tool in high-dimensional dependence modelling. In depth details on vine copula modelling, estimation and inference are available in Aas et al. (2009), Dissmann (2010), Kurowicka and Joe (2011), Dissmann, Brechmann, Czado, and Kurowicka (2013) and Brechmann (2010). In the following paragraphs we summarise the most important facts about vines.

A vine V on n variables is defined as a nested set of connected trees $V = \{T_1, \dots, T_{n-1}\}$, where the edges of tree j are the nodes of tree $j + 1$, $j = 1, \dots, n - 2$. A regular vine on n variables is a vine in which two edges in tree j are joined by an edge in tree $j = 1$ only if these edges share a common node, $j = 1, \dots, n - 2$. Kurowicka and Cooke (2003) provide the following definition of a Regular vine.

Definition 1. (Regular vine). V is a regular vine on n elements with $E(V) = E_1 \cup \dots \cup E_{n-1}$ denoting the set of edges of V if

1. $V = \{T_1, \dots, T_{n-1}\}$,
2. T_1 is a connected tree with nodes $N_1 = \{1, \dots, n\}$, plus edges E_1 ; for $i = 2, \dots, n - 1$, T_i is a tree with nodes $N_i = E_{i-1}$,
3. (proximity) for $i = 2, \dots, n - 1$, $\{a, b\} \in E_i$, $\#a \triangle b = 2$, where $\triangle$ denotes the symmetric difference operator and $\#$ denotes the cardinality of a set.

An edge in a tree T_j is an unordered pair of nodes of T_j or equivalently, an unordered pair of edges of T_{j-1} . By definition, the order of an edge in tree T_j is $j - 1$, $j = 1, \dots, n - 1$. The degree of a node is determined by the number of edges attached to that node. A regular vine is called a *canonical vine*, or *C-vine*, if each tree T_i has a unique node of degree $n - 1$ and therefore, has the maximum degree. A regular vine is termed a *D-vine* if all the nodes in T_1 have degrees no higher than 2.

Definition 2. (Constrained set (Cooke, Joe, Aas, Kurowicka, & Joe, 2011)). For $e \in E_i$, $i \leq n - 1$, the constraint set associated with e is the complete union of U_e^* of e , which is the subset of $\{1, \dots, n\}$ reachable from e by the membership relation.

For $i = 1, \dots, n - 1$, $e \in E_i$, if $e = \{j, k\}$, then the conditioning set associated with e is

$$D_e = U_j^* \cap U_k^*$$

and the conditioned set associated with e is

$$\{C_{e,j}, C_{e,k}\} = \{U_j^* \setminus D_e, U_k^* \setminus D_e\}.$$

Proposition 3. *If $V=T_1, \dots, T_{n-1}$ is a regular vine, then*

1. *the number of edges is $n(n-1)/2$,*
2. *each conditioned set is a doubleton, each pair of variables occurs exactly once as a conditioned set,*
3. *if two edges have the same conditioning set, then they are the same edge.*

Figure-1 gives an example of an R-vine with five dimensions where all edges $e \in E(V)$ are labelled according to the convention $ij|K$ with $i, j \in \{1, \dots, n\}$ and $K \subset \{1, \dots, n\}$. The vertices in tree T_{j+1} are the edges E_j of tree T_j . In figure-1, the edge in T_4 is $V1V3|V2V4V5$.

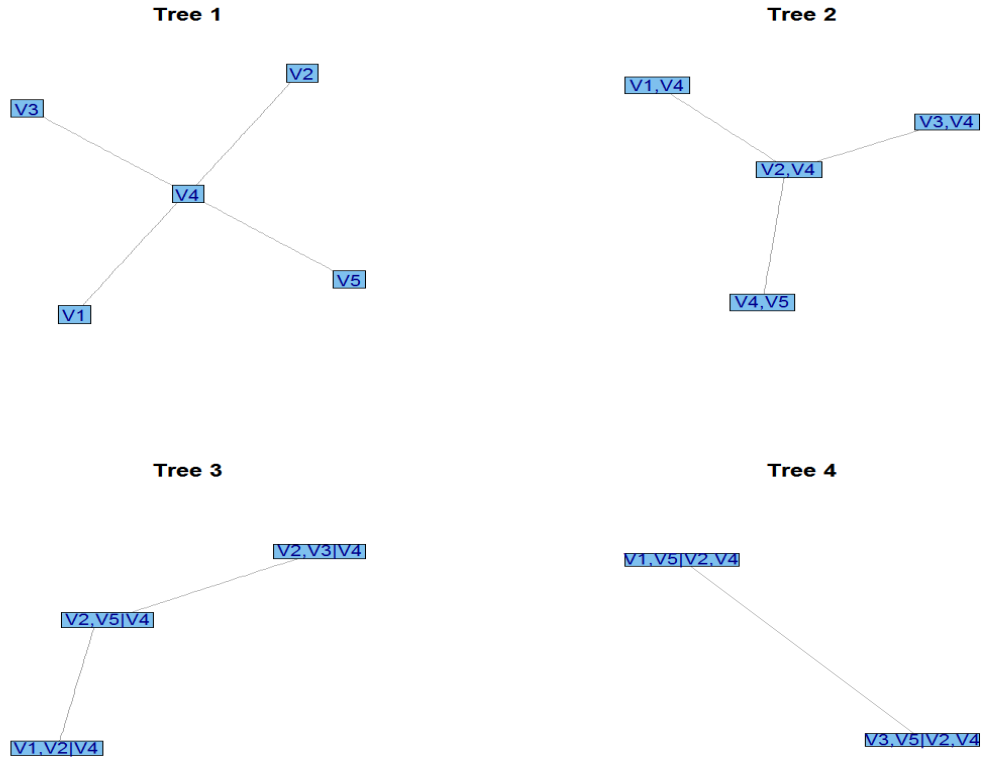


Figure 1: Five-dimensional R-vine

To model R-vine copula one must specify $n - 1$ unconditional bivariate copulas between variables indexed by the conditioned sets of the edges in the first tree of the R-vine. For the second tree of the R-vine one needs to specify the bivariate copulas between variables indexed by the conditioned sets conditional on variables indexed by the conditioning sets of edges of R-vine. The R-vine copula specification matrix can be formally defined as

Definition 4. (R-vine copula specification (Bedford & Cooke, 2002)). (F, V, C) is an R-vine copula specification if

- $F = (F_1, \dots, F_n)$ is a vector of continuous invertible distribution functions,
- V is an n -dimensional R-vine and
- $C = \{C_e | e \in E(V)\}$ is a set of copula with C_e being a bivariate copula, a so-called PCC.

An n -dimensional cdf F of a random vector $X = (X_1, \dots, X_n)$ realises an R-vine copula specification (F, V, C) thus exhibiting R-vine dependence if,

- $X_i \sim F_i$ for all $i \in \{1, \dots, n\}$ and
- $C_{ij|K}$ is the bivariate copula (PCC) that joins $X_i|X_K$ and $X_j|X_K$ for all edges $ij|K \in E(V)$

The density of an R-vine copula is specified by assigning appropriate bivariate copulas to edges of the R-vine, and is equal to the product of conditional and unconditional copulas assigned its edges (Bedford & Cooke, 2001, 2002). All copulas in an R-vine can be of a different type and their parameters can be specified independently from each other, the choice of copulas also influence each other due to the tree structure of R-vines.

Theorem 5. *There is a unique distribution F which realise an R-vine copula (F, V, C) on n elements with density (see Bedford and Cooke (2001) for proof)*

$$f = \prod_{i=1}^n f_i \prod_{ij|K \in E(V)} c_{ij|K}(F_{i|K}, F_{j|K}).$$

For example, if a cdf F satisfies the R-vine specification as given in figure-1, it has the following density factorisation

$$f = \left(\prod_{i=1}^5 f_i \right) c_{14} c_{24} c_{45} c_{34} c_{12|4} c_{25|4} c_{23|4} c_{15|24} c_{35|24} c_{13|245}.$$

Theorem 5 can be simplified to get the density factorisation for C-vine and D-vine copulas (see Aas et al. (2009)).

Matrix Representation of R-vines

R-vine structures can be represented in matrices, which is a short and convenient way of representing high dimensional R-vines. Dissmann (2010), Dissmann et al. (2013) and Kurowicka (2011) recently suggested a method which involves specifying a lower triangular matrix which stores the specification of all the conditional copulas.

For a lower triangular matrix $M = (m_{j,i})_{j,i \in \{1, \dots, n\}}$, the entries in M on and below the diagonal are from some index I with n elements. The elements in every column are distinct. In M , each row proceeding from the bottom represents a tree, the diagonal entry represents the conditioned set and by the corresponding column entry of the row under consideration. The conditioning set is given by the column entries below this row. Similarly, the corresponding parameters and types of copula can be stored in matrices relating to M (Dissmann, 2010). Lets consider the specification matrix for the R-vine copula in figure-1 as an example.

$$M = \begin{pmatrix} 1 & & & & \\ 3 & 3 & & & \\ 5 & 5 & 2 & & \\ 2 & 2 & 5 & 5 & \\ 4 & 4 & 4 & 4 & 4 \end{pmatrix}.$$

In the matrix above, the first column specifies the copulas $c_{13|524}, c_{15|24}, c_{12|4}$ and c_{14} . Similarly it can be seen that this matrix results in the following R-vine copula density factorisation

$$f = \left(\prod_{i=1}^5 f_i \right) c_{14} c_{24} c_{45} c_{34} c_{12|4} c_{25|4} c_{23|4} c_{15|24} c_{35|24} c_{13|245}.$$

We will now discuss the dataset and methodology in the next section.

3 Data & Methodology

We use a dataset of daily log returns of the major stock indices of the eleven Asian financial markets of Thailand, Malaysia, Indonesia, Singapore, Philippine, Korea, Japan, China, Hong Kong, Taiwan, India and major developed international financial markets of US, UK and Australia. Table-1 provides the list of stock indices used in the study along with the abbreviated names. We obtained the daily stock prices in US Dollars from Datastream for two different time periods viz., around the Asian Financial Crisis (AFC) of 1997-1998 and the GFC of 2007-2009. In particular, for the AFC period we study the time periods from January 1995 to December 2000, covering pre-AFC (January 1995 to December 1996), AFC (January 1997 to December 1998) and post-AFC (January 1999 to December 2000). The analysis around the GFC covers the time period from January 2005 to December 2011, covering pre-GFC (January 2005 to June 2007), GFC (July 2007 to August 2009) and post-GFC (September 2009 to December 2011).

Table 1: Stock Indices used in the study

	Country	Stock Index	
1	Japan	NIKKEI 225 STOCK AVERAGE - PRICE INDEX	NIKKEI
2	Indonesia	IDX COMPOSITE - PRICE INDEX	JAKCOMP
3	Thailand	BANGKOK S.E.T. - PRICE INDEX	BNGKSET
4	Singapore	SINGAPORE STRAITS T. DS-CALCULATED - PRICE INDEX	SNGPORZ
5	Phillippines	PHILIPPINE SEI (PSEi) - PRICE INDEX	PSECOMP
6	South Korea	KOREA SE COMPOSITE (KOSPI) - PRICE INDEX	KORCOMP
7	China	SHANGHAI SE COMPOSITE - PRICE INDEX	CHSCOMP
8	Hong Kong	HANG SENG - PRICE INDEX	HNGKNGI
9	Taiwan	TAIWAN SE WEIGHED TAIEX - PRICE INDEX	TAIWGHT
10	India	S&P BSE (SENSEX) 30 SENSITIVE - PRICE INDEX	IBOMSEN
11	Malaysia	FTSE BURSA MALAYSIA KLCI - PRICE INDEX	FBMKL
12	Australia	ASX ALL ORDINARIES - PRICE INDEX	ASXAORD
13	United States	S&P 500 COMPOSITE - PRICE INDEX	SP500
14	United Kingdom	FTSE 100 - PRICE INDEX	FTSE100

Table-2 gives the major descriptive statistics for the daily log returns during the two time periods. The data exhibit non-Gaussian heavy tailed distribution, which is supported by the Shapiro-Wilk test for normal distribution and significant values of Skewness and Excess Kurtosis in table-2. Figure-2, gives the Quantile-Quantile (Q-Q) plots for the data series during the AFC which also show that the price return series are not normally distributed. All the return series in the data show high volatility periods during the AFC and the GFC. Figure-3 shows the return graphs for the indices during the GFC, with all the return graphs showing a high level of volatility, particularly during 2008-2009.

The analysis requires standardised marginals to fit to the vine copula models. We fit the return series to different skewed student-t ARCH/GARCH models in order to obtain the residuals from the model which fit the best to the time series based on its AIC criteria. GARCH (1,1), ARMA (1,0)-GARCH (1,1), ARMA (1,1)-GARCH(1,1) and ARMA (0,1)-GARCH(1,1) are tested to find the best fit. The residuals from the best fit model are then standardised using the standard deviation from the model. The standardised residuals are then converted to uniform marginals using the Rank method. These marginals are then used in the vine copula selection process.

Table 2: Descriptive Statistics for the Data Samples

Sample 1 (1995-2000)														
	NIKKEI	JAKCOMP	BNGKSET	SNGPORZ	PSECOMP	KORCOMP	CHSCOMP	HNGKNGI	TAIWGHT	IBOMSEN	FBMKL	ASXAORD	SP500	FTSE100
N	1563	1563	1563	1563	1563	1563	1563	1563	1563	1563	1563	1563	1563	1563
Min	-6.6665	-38.5662	-14.0712	-8.3734	-10.8941	-19.1055	-17.9442	-14.7132	-11.3456	-8.597	-37.0103	-7.0747	-7.1127	-3.5342
Max	12.5711	22.3144	14.82	10.7123	21.2658	27.4437	26.9362	17.2702	6.7405	9.0162	23.4081	7.3481	4.9887	4.0432
Median	-0.0483	0	-0.1537	0.0122	-0.055	0	0.003	0.0087	-0.0042	0	-0.0164	0.0056	0.0425	0.0621
Mean	-0.0316	-0.1027	-0.1386	0.003	-0.0857	-0.0754	0.0757	0.0386	-0.0406	-0.0283	-0.0483	0.0114	0.0676	0.0423
Std. Dev.	1.6427	3.5069	2.361	1.6007	1.9501	3.0663	2.0905	1.9307	1.7398	1.7218	2.6069	1.0794	1.0627	0.9905
Coef. Var.	-52.0552	-34.1557	-17.0357	528.2573	-22.7436	-40.6621	27.6087	50.0231	-42.8463	-60.7557	-54.001	94.8635	15.7294	23.3955
Skewness	0.4156	-1.0926	0.6008	0.1325	0.8247	0.3153	0.8455	0.1893	-0.2783	0.0499	-1.1043	-0.0228	-0.3742	-0.0338
SE (Skewness)	3.3568	-8.8254	4.8534	1.0702	6.6619	2.5472	6.8298	1.5294	-2.2478	0.4031	-8.9202	-0.1843	-3.0223	-0.273
Kurtosis	3.7633	20.0603	5.2277	4.2389	13.3192	10.587	24.054	9.9233	3.6623	2.4506	40.613	4.2104	4.5817	1.211
SE (Kurtosis)	15.209	81.0726	21.1273	17.1314	53.8288	42.7866	97.2129	40.1044	14.801	9.9038	164.1351	17.0162	18.5165	4.8944
Shapiro-Wilk Test	0.968	0.759	0.9291	0.9528	0.8906	0.8681	0.8441	0.9074	0.9541	0.9742	0.7352	0.9667	0.9532	0.9877
P-Value	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Sample 2 (2005-2011)														
	NIKKEI	JAKCOMP	BNGKSET	SNGPORZ	PSECOMP	KORCOMP	CHSCOMP	HNGKNGI	TAIWGHT	IBOMSEN	FBMKL	ASXAORD	SP500	FTSE100
N	1825	1825	1825	1825	1825	1825	1825	1825	1825	1825	1825	1825	1825	1825
Min	-11.1856	-13.815	-16.0642	-8.6282	-13.9096	-20.4794	-9.1014	-13.5888	-7.1712	-11.9071	-11.0144	-15.8491	-9.4695	-10.5381
Max	11.6442	12.8617	9.6726	10.349	9.3331	24.64	9.0194	13.4043	8.2571	19.0515	5.0287	8.1092	10.9572	12.2189
Median	0.0141	0	0	0.0639	0.0456	0	0.0287	0.0267	0.0484	0.0244	0.0211	0.0831	0.0544	0.0609
Mean	-0.0011	0.0747	0.0349	0.032	0.0615	0.0327	0.0452	0.0142	0.0102	0.0356	0.0386	0.0155	0.002	-0.0036
Std. Dev.	1.5795	1.8634	1.5599	1.4374	1.576	2.1338	1.8055	1.7577	1.4903	1.9637	1.0257	1.8711	1.4363	1.666
Coef. Var.	-1441.3452	24.9525	44.7369	44.9157	25.6458	65.2624	39.9036	123.5455	145.5063	55.1113	26.5847	120.8529	708.4821	-466.5937
Skewness	-0.353	-0.6045	-1.0113	-0.2286	-0.6942	-0.2593	-0.3335	0.0607	-0.2884	0.1668	-0.8537	-0.9435	-0.2886	-0.0903
SE (Skewness)	-3.081	-5.276	-8.826	-1.995	-6.0586	-2.2631	-2.9104	0.5302	-2.5169	1.4559	-7.4503	-8.2347	-2.5183	-0.7884
Kurtosis	6.6133	7.6878	11.7138	5.7977	6.4676	17.9549	3.1924	8.4007	2.9327	7.3453	9.4426	7.8574	9.3676	8.2872
SE (Kurtosis)	28.8742	33.5654	51.1432	25.313	28.238	78.392	13.9381	36.678	12.8045	32.0701	41.2271	34.3058	40.8995	36.1825
Shapiro-Wilk Test	0.9356	0.9012	0.9071	0.9292	0.9416	0.8664	0.9507	0.9046	0.952	0.935	0.9303	0.9174	0.8745	0.902
P-Value	0	0	0	0	0	0	0	0	0	0	0	0	0	0

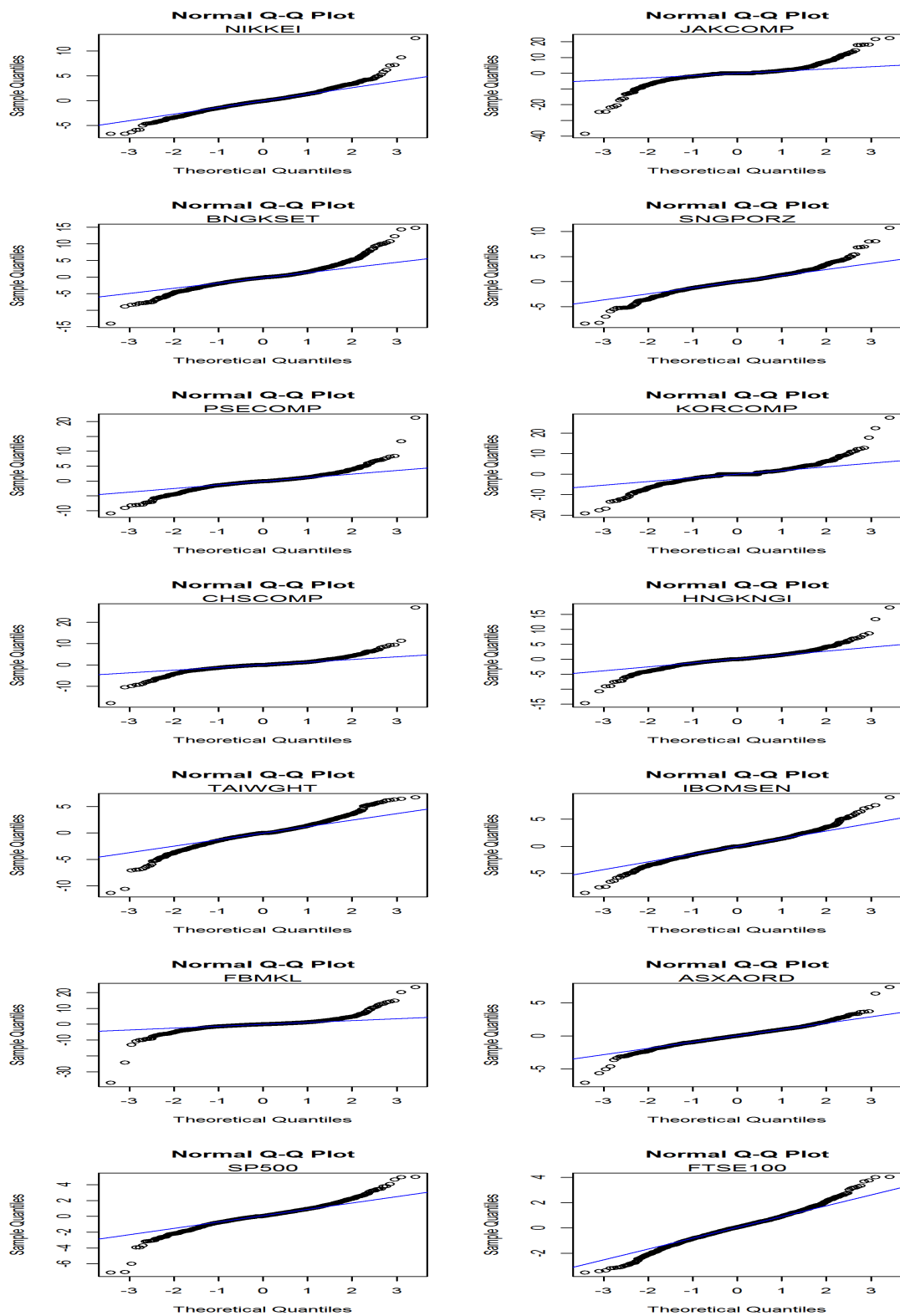


Figure 2: Q-Q Plots for return data covering AFC

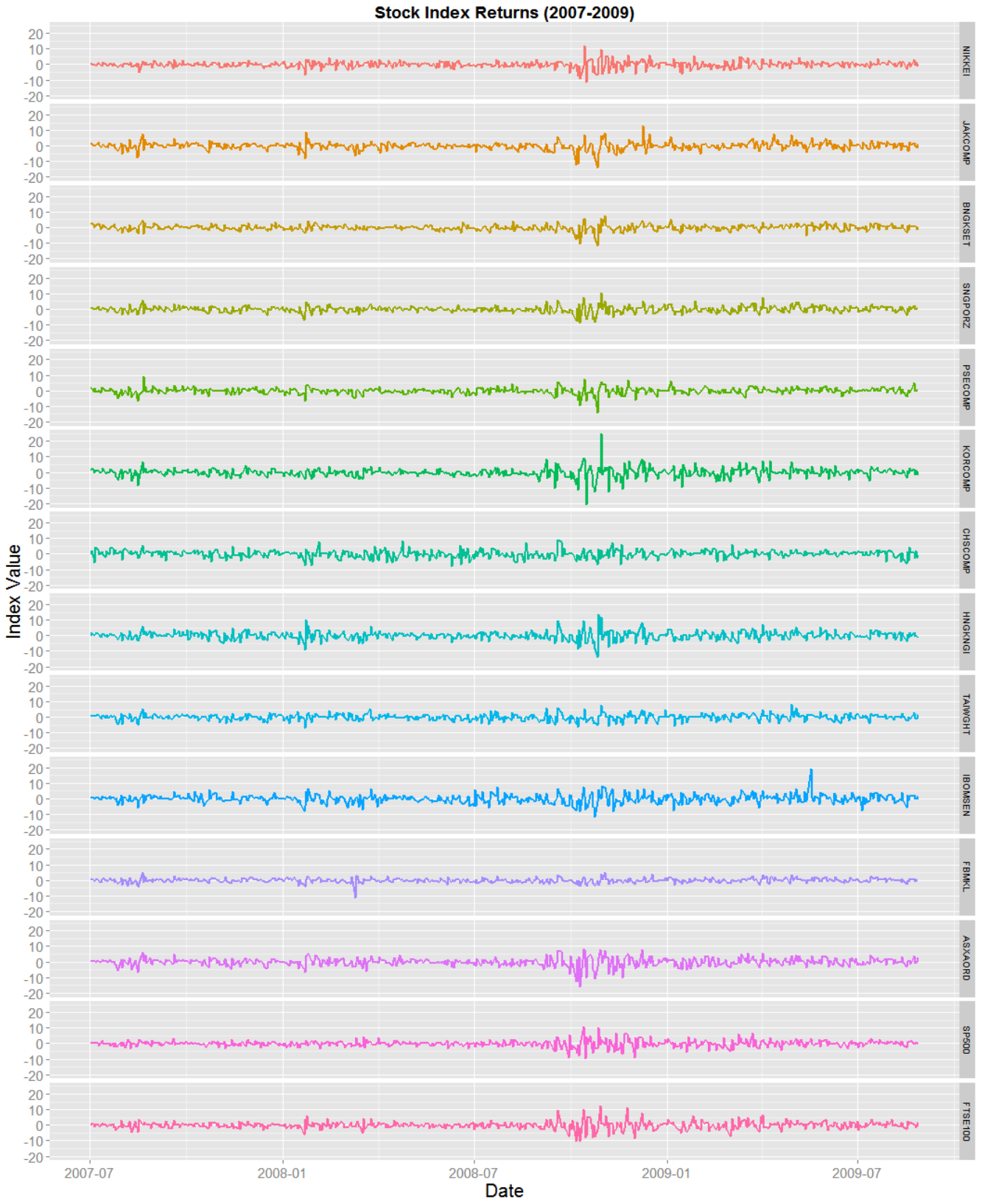


Figure 3: Return Plot (2007-2009)

Vine copulas provide the flexibility of various copulas to select from for the construction of the vine tree. In this analysis we use the range of copulas from 1 to 6; 1 = Gaussian copula, 2 = Student t copula (t-copula), 3 = Clayton copula, 4 = Gumbel copula, 5 = Frank copula, 6 = Joe copula. The copulas are selected using the AIC criterion. R package VineCopula has been used to fit the vine copula models. R-vine trees are selected using maximum spanning trees with absolute pairwise Kendall's taus as weights. The selection is done by optimising the following for each tree:

$$\max \sum_{e_{ij} \in T_s} |\hat{\tau}_{ij}|,$$

where edges e_{ij} are in the spanning tree (tree on all nodes) T_s and $\hat{\tau}_{ij}$ denote the pairwise empirical Kendall's tau. The setting of the first tree selection step is always a complete graph. The proximity condition principle in R-vine construction defines the setting for the subsequent trees (Brechmann, Czado, & Aas, 2012; Dissmann et al., 2013). In C-vine the root node is selected in a similar way by identifying the node with the strongest dependencies to all other nodes. Computationally this is done by taking the node with the maximum column sum in the empirical Kendall's tau matrix.

We will now present the major results from the analysis in the next section, starting with the results from the AFC.

4 Results

The results are presented in two parts. In the first part we discuss the dependence structure obtained from R-vine copula during the period of the AFC. The second subsection will discuss the dependence structure from the GFC time period.

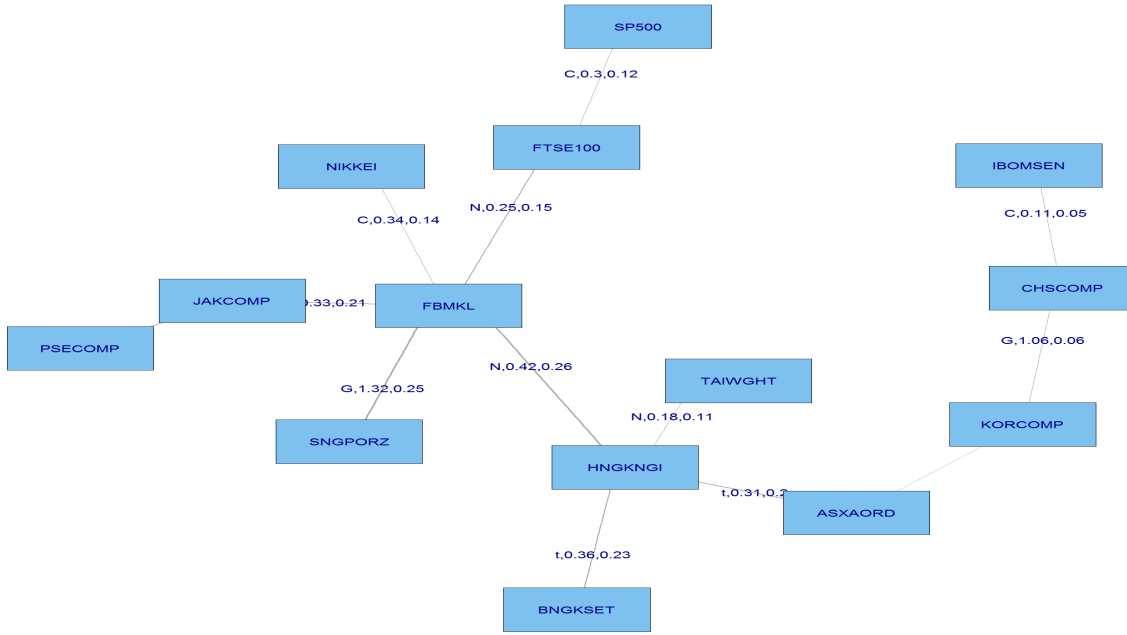
4.1 Dependence during the AFC

We model the dependence structure among the Asian markets and the international markets in our data sample around the AFC (January 1995 to December 2000). The analysis covers pre-AFC (January 1995 to December 1996), AFC (January 1997 to December 1998) and post-AFC (January 1999 to December 2000).

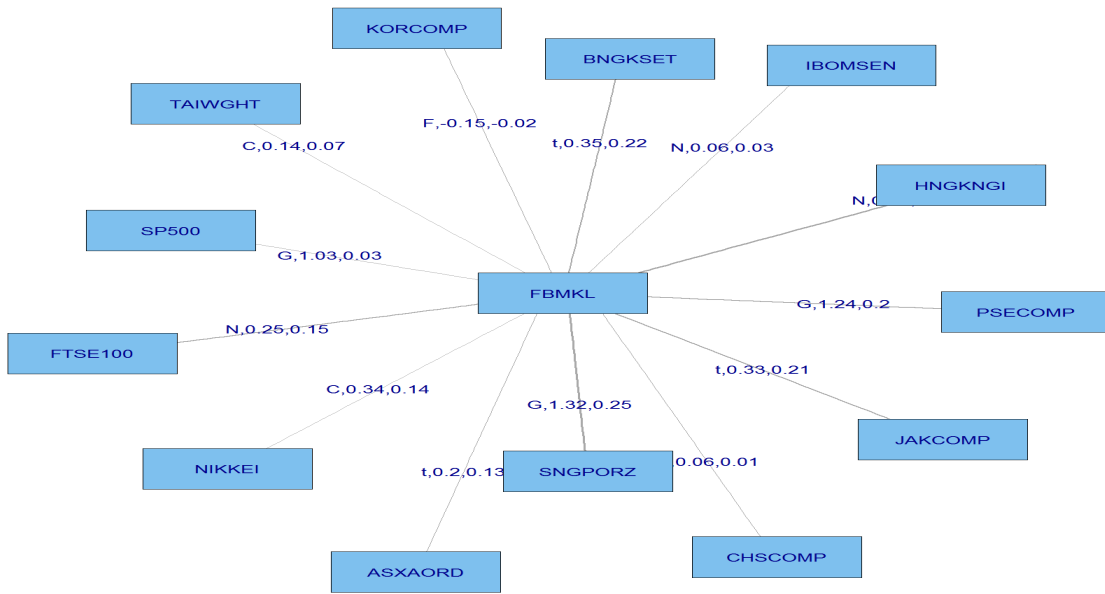
R-vine dependence during the pre-AFC period (1995-1996)

Although the empirical analysis has used R-vine copulas, figure-4 (b), shows the first tree for C-vine selection of the pre-AFC data sample for illustration. C-vine structures have a limitation of a unique root node that should be connected to all other nodes, which can be seen in figure-4(b) where the Malaysian stock index has the highest dependence with all the other indices and is the root node connected to all other markets. R-vine copulas are more flexible and do not suffer from this limitation as shown in figure-4(a), where we can see two major cliques formed with the central nodes of Malaysia and Hong Kong. We will concentrate on the results of the R-vine analysis from

here on. Majorly we will focus on examining the change (if any) in dependencies among the stock indices in our data sample during the AFC.



(a) R-Vine



(b) C-Vine

Figure 4: R-Vine and C-Vine Tree 1, Pre-AFC Period

It is evident from figure-4(a) and table-3, that R-vine copulas are more flexible and present a better dependence structure compared to the C-vine copulas. From table-3 it can be seen that majority of the markets are dependent on the market of Hong Kong (8) and Malaysia (11). However, as seen from the second column in table-3, Phillippines (5) had the highest degree of dependence with Indonesia (2) during the pre-AFC period. Similarly there are other index pairs showing different dependence during the pre-AFC period, e.g. US (13) and UK (14) from column-3, which makes an intuitive dependent pair being the largest financial markets of the world. Table-4 gives the type of copulas fitted to the different bivariate pairs (conditional or unconditional) in the R-vine specification. Following the US and UK relationship from column-3 in table-3, table-4 shows that the best fit copula in this case is the Clayton copula (3), which shows that these two markets are positively related. The Clayton copula parameter value in this case is 0.303 as given in table-5(a). The estimated parameter value for the copulas in table-4 are given in table-5 (a) and (b). Table-5(b) gives the second parameter for the copulas which are defined using two parameters in the copula selection set. Here Student-t and Frank copula have two parameters.

Table 3: R-Vine Copula Specification Matrix (Pre-AFC)

9													
13	5												
14	13	13											
10	14	2	14										
7	10	3	2	10									
6	7	4	3	2	7								
12	6	1	4	3	2	2							
1	12	10	1	4	3	6	3						
5	1	7	10	1	4	12	6	4					
2	4	6	7	11	1	1	12	6	1				
4	3	12	6	8	11	4	1	12	6	11			
11	8	8	12	12	8	3	4	1	12	6	6		
3	11	11	8	6	12	8	11	8	8	12	8	12	
8	2	14	11	7	6	11	8	11	11	8	12	8	8

Once these initial nodes and edges have been taken into account, the second row is conditioned on their dependency relationship, and is the next branch in the tree. In the first entry at the top of the third column in table-3, we have index 13 , UK, but when its relationship with the US is taken into account (last entry in the column), the penultimate entry in that column is index 11, Malaysia, indicating, quite surprisingly, that there is a co-dependency relationship between these two pairs of indices, after conditioning on the US market. Similarly, complex patterns of dependency can be readily captured. It can be seen that at different dependencies conditioned across the same node six different copulas are used. This variety of usage is apparent across table-4 at various levels in the Tree structures used to capture dependencies.

Table 4: R-Vine Copula Type Matrix (Pre-AFC)

1
1 3
5 1 3
3 5 5 5
1 5 5 6 1
5 2 3 1 1 5
1 5 5 3 2 5 5
1 1 5 3 5 3 4 3
4 5 2 5 3 6 5 5 5
5 2 3 1 1 3 4 3 5 2
5 3 3 5 1 3 3 3 4 5 5
5 5 1 5 5 1 1 1 5 1 4 5
1 4 3 1 3 4 2 2 4 3 1 1 2

Table 5: R-Vine Copula Parameters (Pre-AFC)

(a) Copula Parameter-1												(b) Copula Parameter-2											
0.067												0											
-0.0368	0.1018											0 0											
0.1887	-0.0417	0.1483										0 0	0										
0.0825	0.0702	0.1348	0.2912									0 0	0	0									
-0.0253	-0.2023	-0.2126	1.0412	0.0815								0 0	0	0 0									
0.1029	-0.0395	0.0152	0.0035	-0.0103	0.1127							0 11.6347	0	0 0	0								
-0.0114	0.505	-0.2059	0.0555	-0.0258	0.1868	-0.2475						0 0	0	0 10.9581	0 0								
0.0539	0.0168	-0.2872	0.0614	-0.5276	0.0345	1.027	0.0217					0 0	0	0 0	0 0	0							
1.0166	0.6289	0.0368	-0.1053	0.0552	1.0423	-0.1378	-0.1426	-0.1723				0 0	5.6907	0 0	0 0	0	0						
0.5292	0.0703	0.0578	0.0542	0.0161	0.0475	1.0466	0.0387	0.2371	-0.025			0 7.7191	0	0 0	0 0	0	0 8.8361						
0.4115	0.1309	0.0924	0.9918	0.0345	0.0886	0.2075	0.233	1.0625	0.4198	-0.3417		0 0	0	0 0	0 0	0	0 0	0					
-0.3457	1.3403	0.0071	0.7904	-0.053	-0.0434	0.2255	0.248	0.805	0.1408	1.0534	0.1458	0 0	0	0 0	0 0	0	0 0	0 0					
0.181	1.3044	0.303	0.2472	0.1087	1.0639	0.3288	0.3634	1.3178	0.337	0.4182	0.1079	0.3076	0 0	0	0 0	0 6.7624	7.0119	0 0	0 0	9.5929			

It can be seen from the copula parameters in table-5 that most of the dependencies modelled by R-vine copulas are positive dependencies. This shows that before the AFC these markets mostly had positive dependence which was true for all the unconditional dependencies. In the next stage of our analysis we examine if these dependencies changed during the AFC.

R-vine dependence during the AFC period (1997-1998)

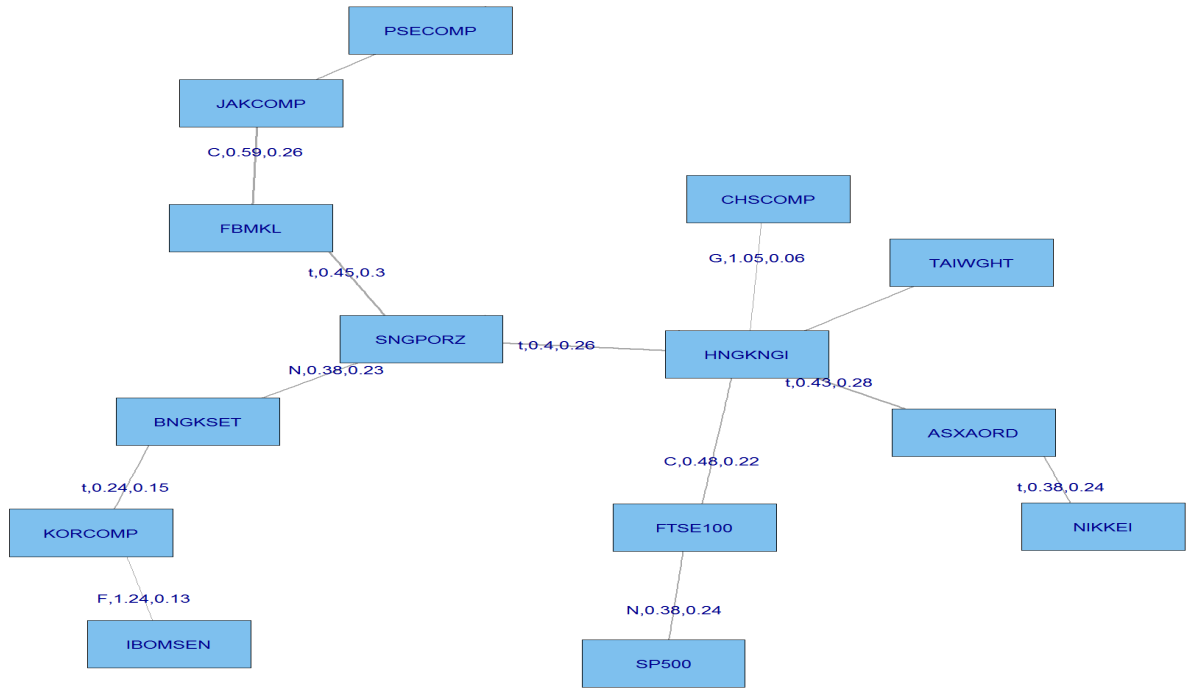
Figure-5, gives the 1st and 2nd trees of the fitted R-vine copulas to the stock indices returns during the AFC period. When compared to figure-4, the change in the dependence structure is apparent due to the impact of the AFC. For example, the markets of Japan, UK, Singapore and US had higher dependence with the market of Malaysia which appears to have changed during the AFC period. In this case Japan, UK, US have higher dependence with Hong Kong. The change in groupings in the

R-vine tree reflects the change in the financial markets during the AFC. Indonesia, Malaysia and the Phillippines are grouped together and were among the most severely affected countries during the AFC. Two other countries which were quite severely affected are Thailand and South Korea which are also grouped together.

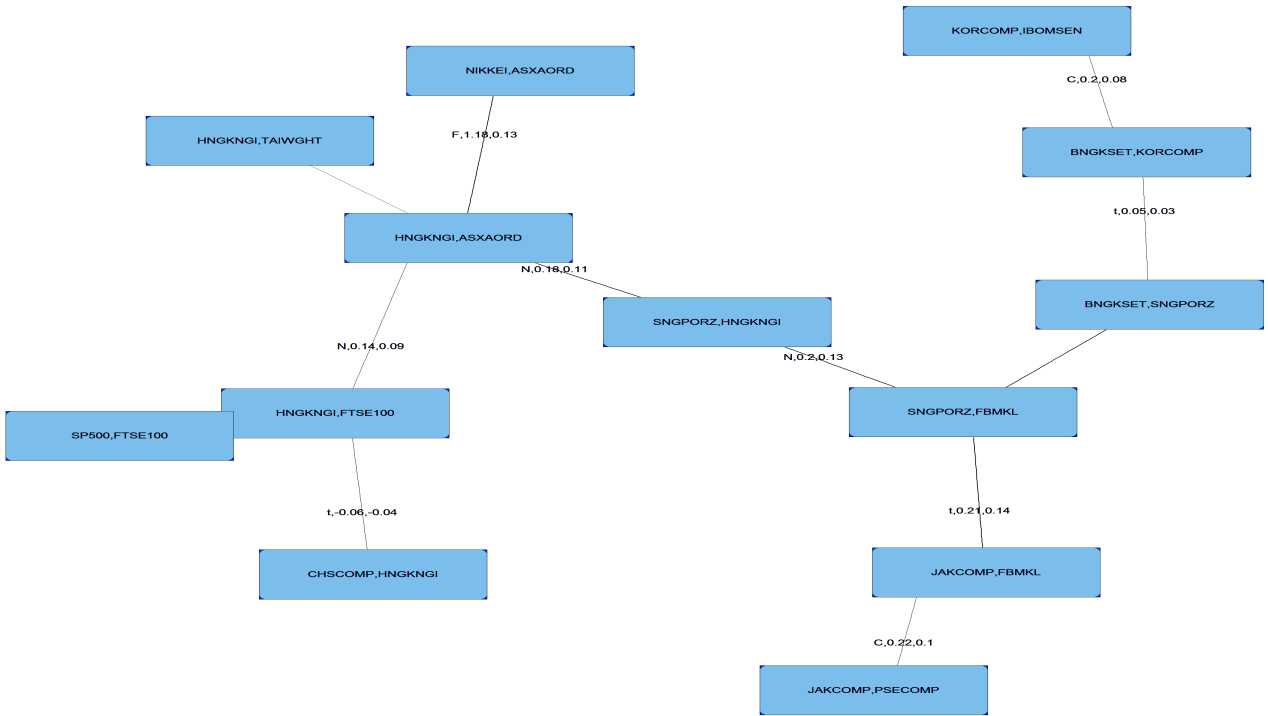
Table-6 gives the detailed R-vine structure matrix followed by table-7 with the selected copulas in the structure for this period. The dependence structure now changes dramatically as a result of the AFC. It can be seen that in general the developed markets of Australia (12), US (13) and UK (14) showed less dependence with the worst hit south east Asian countries during AFC. With a considerable change in the copulas selected, i.e, an increase in copulas with tail dependence (2, 3, 4 and 6) table-7 also shows that there was also a change in the joint tail behaviour of these markets. This is not surprising, as we would expect the tails of the distributions to increase during periods of financial distress. The estimated copula parameters for the dependence are given in table-8.

Table 6: R-Vine Copula Specification Matrix (AFC)

10													
5	5												
2	13	2											
13	7	13	6										
7	14	7	13	9									
14	1	14	7	13	3								
1	9	1	14	7	13	11							
9	12	9	1	14	7	13	4						
12	8	12	9	1	14	7	13	13					
8	6	8	12	3	1	14	7	1	7				
11	3	6	8	11	12	1	14	12	1	1			
4	4	3	11	4	8	12	1	7	12	14	12		
3	11	4	4	12	11	8	12	8	14	8	14	14	
6	2	11	3	8	4	4	8	14	8	12	8	8	8



(a) R-Vine (Tree-1)



(b) R-Vine (Tree-2)

Figure 5: R-Vine Tree 1 &2 , AFC Period

Table 7: R-Vine Copula Type Matrix (AFC)

6
2 5
1 5 1
4 3 5 5
6 5 1 2 1
5 1 3 5 2 3
1 5 4 6 5 3 5
6 5 5 5 6 6 5 5
3 5 4 1 1 4 1 2 5
3 5 6 4 3 1 2 4 3 1
3 3 1 3 1 2 3 1 2 5 4
3 3 2 2 1 2 1 1 2 2 5 1
5 1 3 2 2 1 2 2 1 4 2 2 3

Table 8: R-Vine Copula Parameters (AFC)

(a) Copula Parameter-1

1.017
-0.0039 0.6963
-0.0107 0.2331 -0.0496
1.0109 0.047 0.1864 0.4018
1.046 -0.3411 0.0216 -0.0205 -0.0508
-0.2096 0.0321 0.0633 -0.0992 -0.0027 0.0583
-0.0412 0.6739 1.028 1.0485 -0.349 0.07 0.1249
1.0142 1.0824 1.1779 0.5132 1.0669 1.0229 -0.5984 -0.2912
0.0729 0.2785 1.0454 0.0592 0.0713 1.0328 0.0088 -0.0375 -0.1992
0.0946 1.1564 1.107 1.0727 0.0678 0.1087 0.109 1.0183 0.0884 -0.0742
0.0579 0.2579 0.1539 0.1218 0.0983 0.1571 0.1617 0.1462 -0.0675 0.1886 1.07
0.1983 0.2158 0.208 0.0506 0.1387 0.2151 0.1973 0.1757 0.0609 -0.0592 1.1845 0.1358
1.236 0.3884 0.5934 0.2411 0.2431 0.3841 0.4533 0.4017 0.3772 1.0545 0.3819 0.4297 0.4789

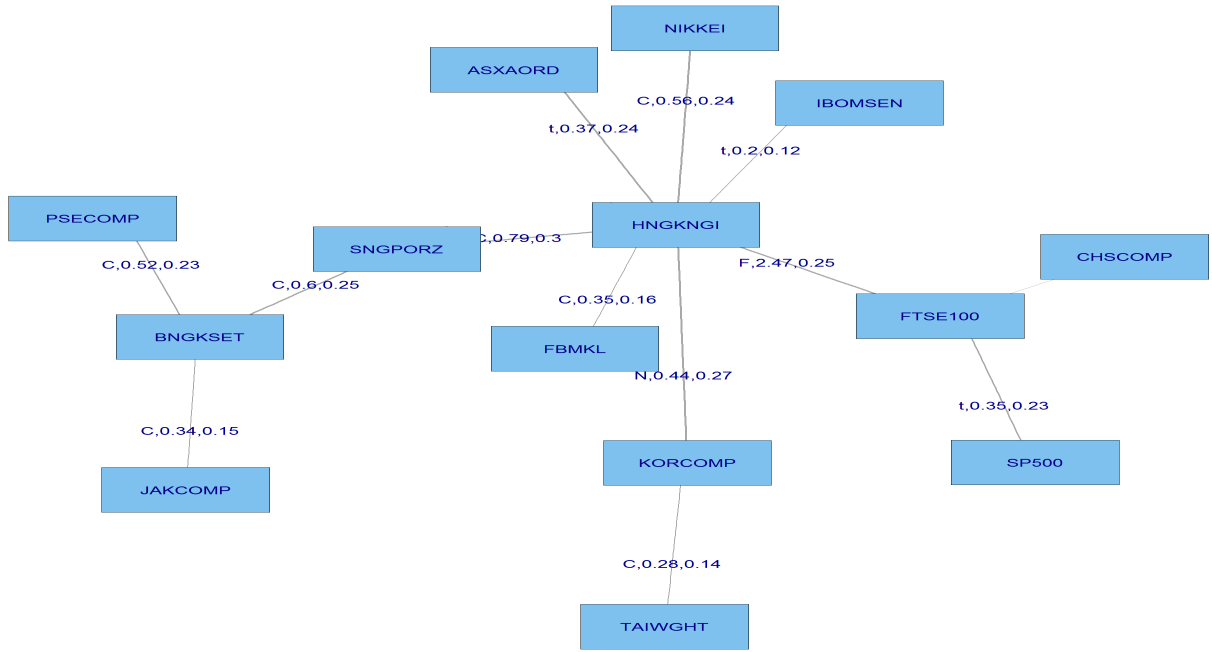
(b) Copula Parameter-2

0
12.967 0
0 0 0
0 0 0 0
0 0 0 13.8456 0
0 0 0 0 7.0987 0
0 0 0 0 0 0 0
0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 12.8073 0
0 0 0 0 0 0 11.8986 0 0 0
0 0 0 0 0 8.5078 0 0 11.2804 0 0
0 0 5.934 7.3498 0 10.7593 0 0 10.3475 6.6719 0 0
0 0 0 13.8802 5.2416 0 6.2904 9.3886 0 0 5.648 7.6583 0

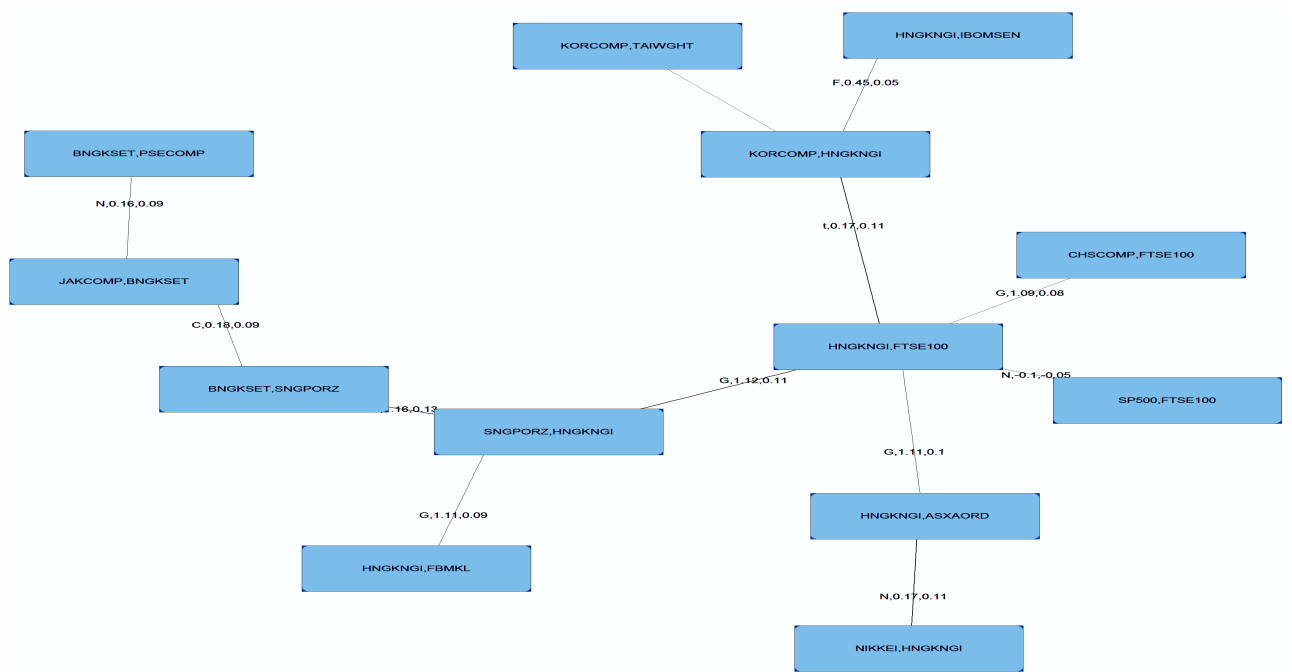
R-vine dependence during the post-AFC period (1999-2000)

The final segment in the analysis of the AFC is for the post-AFC period, 1999 to 2000. Figure-6, gives the first two trees of the R-vine structure during the post-AFC period. The change in the dynamics of these stock markets is apparent in the new structure followed by the AFC. It is evident that the international markets have continued to be more dependent on Hong Kong which has also attained higher dependence with the markets of India and Malaysia which moved away from Singapore. A shift in the dependence of the Chinese stock market can also be seen in the post-AFC period, which shows higher dependence with the UK market as compared to Hong Kong in the AFC

period, and Korea and India in the pre-AFC period. This could be a result of growth in Chinese economy and its influence.



(a) R-Vine (Tree-1)



(b) R-Vine (Tree-2)

Figure 6: R-Vine Tree 1 & 2 , post-AFC Period

Table-9 and table-10 give the R-vine specification matrix and selected copula matrix respectively. Table-10 shows that not only the structure of dependence (table-9) has changed after the AFC but also the type of dependence has evolved. The last row in table-10 shows an increase in Clayton copula (3), implying that during the post-AFC period there was a positive dependence in the left tail of the markets. Similarly are return to normal market conditions can be observed with an increase in dependence defined by Gaussian copula (1) and Frank copula (5) which have no tail dependence. The estimated parameters for the copulas are given in table-11.

Table 9: R-Vine Copula Specification Matrix (post-AFC)

10													
11	13												
7	11	9											
1	7	11	6										
12	1	7	11	11									
5	12	1	7	7	5								
2	5	12	1	5	7	2							
3	2	5	12	2	1	7	3						
13	3	2	5	1	12	1	7	4					
4	9	3	2	12	14	12	1	7	1				
14	6	4	3	14	8	14	12	1	7	12			
9	4	14	4	3	4	8	14	12	14	7	7		
6	8	8	14	4	2	4	8	14	12	14	8	14	
8	14	6	8	8	3	3	4	8	8	8	14	8	8

Table 10: R-Vine Copula Type Matrix (post-AFC)

6													
5	5												
4	1	5											
3	5	5	5										
5	1	3	5	1									
1	5	1	6	5	5								
1	5	1	5	2	4	5							
3	2	5	1	4	1	1	3						
6	4	4	4	6	1	5	5	1					
1	1	3	1	2	1	1	5	3	3				
5	2	1	3	1	4	5	5	4	5	3			
5	1	5	2	4	1	3	4	4	1	4	4		
2	2	3	1	3	3	3	3	3	3	2	1	5	

Table 11: R-Vine Copula Parameters (post-AFC)

(a) Copula Parameter-1												(b) Copula Parameter-2																	
1.0442															0														
-0.199	0.1554														0	0													
1.0157	-0.0513	0.1958													0	0	0												
0.0361	-0.3003	0.4402	0.099												0	0	0	0											
0.2485	0.0117	0.087	0.1443	-0.0349											0	0	0	0	0										
-0.0135	-0.0711	0.0147	1.1013	0.2057	0.3592										0	0	0	0	0	0									
0.0292	-0.203	0.1114	-0.0563	-0.0271	1.0514	-0.1679									0	0	0	0	10.3584	0	0								
0.0706	0.0143	-0.3382	0.1033	1.0399	0.1038	0.0483	0.0421								0	9.5116	0	0	0	0	0	0							
1.0535	1.0305	1.041	1.0154	1.0876	0.0713	0.3804	0.6482	0.0262							0	0	0	0	0	0	0	0							
0.01	-0.0268	0.1496	0.1569	-0.0257	0.104	-0.0989	0.7661	0.1111	0.0439						0	0	0	0	11.9984	0	0	0	0	0					
0.4846	0.049	-0.0685	0.1991	0.1275	1.0538	0.5974	0.2088	1.0772	-0.0833	0.1406					0	11.2084	0	0	0	0	0	0	0	0					
0.4455	-0.1014	0.6713	0.1682	1.1072	0.1566	0.1788	1.1591	1.117	0.17	1.108	1.0931				0	0	0	9.7809	0	0	0	0	0	0	0		0		
0.2027	0.3545	0.2849	0.4413	0.3533	0.5215	0.3399	0.604	0.7928	0.5624	0.3734	-0.1349	2.4655			10.9242	11.5882	0	0	0	0	0	0	0	10.4228	0	0			

4.2 Dependence Analysis During the GFC

In this second part of the analysis, we will now focus on the GFC to analyse the dependence structure of these markets. The GFC was yet another turbulent period which hit almost all the financial markets globally. We model the dependence structure among the Asian markets and the international markets in our data sample around the GFC (January 2005 to December 2011) covering pre-GFC (January 2005 to June 2007), GFC (July 2007 to August 2009) and post-GFC (September 2009 to December 2011). We will first present the results from the pre-GFC period.

R-vine dependence during pre-GFC period (2005-2007)

Figure-7 gives the first two trees from the R-vine structure obtained from the data from the pre-GFC period. Tree-1 in figure-7 shows three different cliques before the onset of the GFC. The developed markets of Australia, US, UK and Japan are in one group with the Philippines showing dependence with Australia. The markets of Thailand, Indonesia, Malaysia, China show higher dependence with Singapore and the markets of Korea, India and Taiwan show higher dependence with Hong Kong. This shows the complex dependence structure of these markets. Tree-1 in figure-7 also shows that Australia-UK, UK-US, and Australia-Japan were all positively dependent on each other.

Table-12 and table-13 give the copula specification and selected copula type matrices for the R-vine copulas. The last column in table-12 gives the copulas fitted as in tree-1 (figure-7), which are mostly Student-t copulas suggesting heavy tails. The estimated parameters for the fitted copulas are given in table-14. If compared to the dependencies during the AFC, it can be observed that the dependence structure has changed since the AFC. We will now analyse how this dependence structure changes during the GFC.

Table 12: R-Vine Copula Specification Matrix (pre-GFC)

7														
13	11													
14	13	2												
11	14	13	13											
2	5	14	10	14										
5	3	5	9	10	5									
3	1	3	6	9	10	3								
1	9	1	1	6	9	10	10							
9	12	9	8	1	6	9	1	9						
10	6	12	5	8	1	6	9	1	6					
12	10	6	3	5	8	1	12	12	1	8				
6	8	10	4	3	3	8	6	4	12	1	1			
8	2	8	12	4	4	12	4	8	4	12	4	12		
4	4	4	14	12	12	4	8	6	8	4	12	4	4	

Table 13: R-Vine Copula Type Matrix (pre-GFC)

3														
1	1													
5	1	2												
1	4	5	3											
1	2	1	1	4										
1	1	3	3	5	3									
5	4	6	5	5	1	1								
5	3	5	5	3	2	3	3							
5	1	3	5	2	1	6	3	5						
6	3	4	5	1	1	5	4	5	5					
5	3	3	5	3	3	1	1	2	3	1				
1	5	1	1	2	2	5	5	5	5	2	2			
2	2	2	2	1	2	2	1	1	2	2	2	2		

Table 14: R-Vine Copula Parameters (pre-GFC)

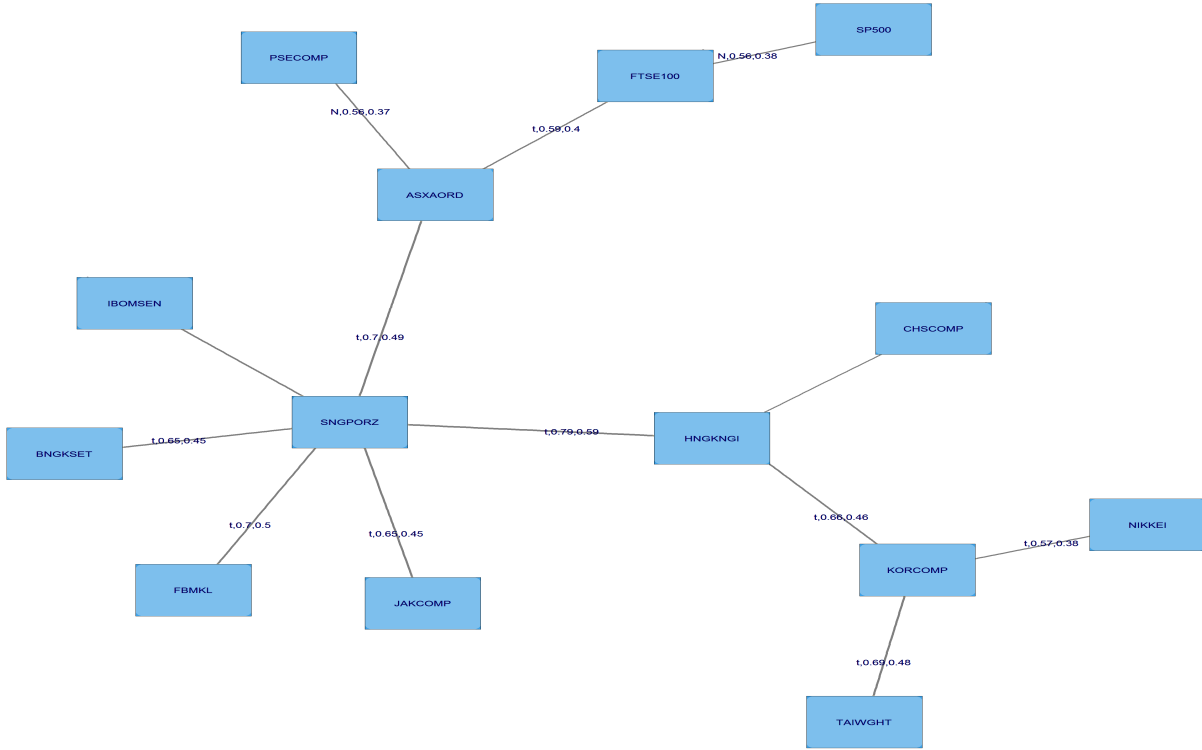
(a) Copula Parameter-1													(b) Copula Parameter-2														
0.056													0														
-0.0089	-0.0268												0	0													
0.4532	-0.0177	-0.0301											0	0	13.7592												
0.0065	1.0897	0.1358	0.0418										0	0	0	0											
-0.0702	0.103	0.1369	-0.0054	1.062									0	9.1119	0	0	0										
-0.0289	-0.0232	0.2123	0.0672	-0.3384	0.0427								0	0	0	0	0	0									
-0.1753	1.0583	1.0492	-0.0816	-0.1227	0.098	0.0206							0	0	0	0	0	0	0								
-0.0964	0.1223	0.4637	0.5515	0.0547	0.0792	0.0886	0.0381						0	0	0	0	10.5214	0	0								
-0.1607	0.0789	0.0859	-0.0579	0.0076	0.0244	1.0486	0.0717	0.6644					0	0	0	0	8.9811	0	0	0	0						
1.0284	0.0508	1.0132	0.0975	-0.046	0.0405	0.1776	1.0414	0.2258	1.4855				0	0	0	0	0	0	0	0	0						
0.3692	0.123	0.1813	0.1928	0.2195	0.1611	0.0937	0.1243	0.1974	0.2124	0.2142			0	0	0	0	0	0	0	15.856	0	0					
0.1038	1.0893	0.2233	-0.1324	0.1542	0.2088	1.2132	1.1594	1.926	1.7731	0.2499	0.3224			0	0	0	0	12.5772	13.4199	0	0	0	13.2956	9.6068			
0.1881	0.496	0.5254	0.3777	0.4904	0.3597	0.4121	0.4252	0.5626	0.5493	0.5915	0.5607	0.544	8.4142	6.1207	6.1468	12.9926	0	11.425	6.7097	0	0	9.1148	9.149	5.24	7.3257		

R-vine dependence during the GFC period (2007-2009)

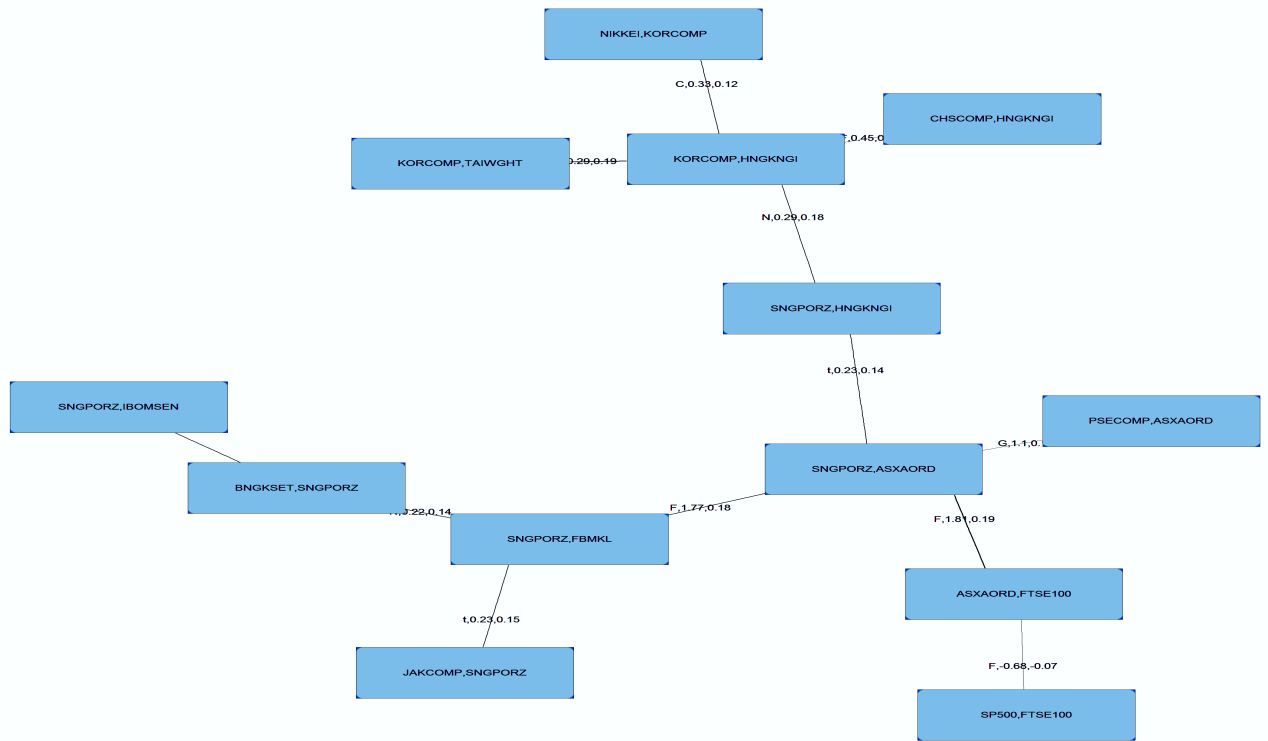
Figure-8 gives the first two trees for the R-vine fit to the data from the GFC period. Tree-1 shows almost the same structure as the pre-GFC period with a few important exceptions. Japan (Nikkei) is no longer in the group with Australia, US, UK and the Phillippines but during this period it shows higher dependence with South Korea and Hong Kong (conditional dependence (tree-2 in figure-8)). The other important shifts in the structure are India-Singapore and China-Hong Kong.

Table 15: R-Vine Copula Specification Matrix (GFC)

10														
7	3													
1	7	2												
9	1	7	7											
6	9	1	13	1										
13	6	9	14	13	9									
14	13	6	1	14	13	6								
8	14	13	9	11	14	13	11							
5	8	14	11	5	11	14	13	8						
12	5	8	5	12	5	11	14	13	5					
2	12	5	12	4	12	5	8	14	13	4				
11	2	12	4	9	4	12	5	5	14	13	12			
3	11	11	6	8	8	4	12	12	4	14	13	14		
4	4	4	8	6	6	8	4	4	12	12	14	13	13	



(a) R-Vine (Tree-1)



(b) R-Vine (Tree-2)

Figure 8: R-Vine Tree 1 & 2 , GFC Period

The other conditional dependencies have also changed during this period as shown in table-15 and table-16. It can be noted that there are more heavy tailed copulas fitted during the GFC period than the pre-GFC period (table-13). This result is consistent with the AFC period where there was also an increase in copulas with heavy tail dependence. The parameter estimates for the copulas are given in table-17.

Table 16: R-Vine Copula Type Matrix (GFC)

3
5 2
5 3 1
1 5 5 5
5 3 6 1 5
1 3 5 5 5 5
1 5 3 5 1 1 2
3 2 5 3 5 6 1 1
5 5 5 3 2 3 3 4 5
3 4 2 5 6 1 3 2 5 5
5 3 3 1 1 4 5 5 2 5 2
5 1 2 5 3 2 1 5 2 4 5 5
1 2 2 1 2 2 2 2 2 1 2 2 1

Table 17: R-Vine Copula Parameters (GFC)

(a) Copula Parameter-1

0.1101
-0.4808 -0.0048
-0.2363 0.0602 -0.051
-0.0246 0.1709 -0.4535 -0.7188
0.3847 0.0454 1.0496 -0.039 -0.8619
0.1661 0.1034 0.0554 -0.2187 -1.7313 -0.1624
0.1487 1.0004 0.0229 -0.2307 -0.0696 -0.0231 0.0529
0.0639 0.14 0.1352 0.0831 0.6269 1.0784 -0.0469 -0.0667
0.4159 -0.0198 0.91 0.0826 0.2274 0.1738 0.1375 1.0183 0.226
0.0622 1.0262 0.1669 -0.3775 1.0351 0.1373 0.1531 0.103 -0.4164 -0.1363
0.5962 0.2197 0.1749 -0.089 0.1216 1.111 1.9515 1.5102 0.202 -0.6144 0.0393
1.1892 0.2202 0.2252 0.446 0.3281 0.2908 0.2876 1.773 0.2327 1.097 1.806 -0.675
0.6495 0.6471 0.655 0.5037 0.5668 0.6946 0.6599 0.7021 0.7925 0.555 0.7016 0.5916 0.561

(b) Copula Parameter-2

0
0 11.8045
0 0 0
0 0 0 0
0 0 0 0 0
0 0 0 0 0 0
0 0 0 0 0 0 10.9766
0 12.5673 0 0 0 0 0 0
0 0 0 0 9.6197 0 0 0 0
0 0 5.5864 0 0 0 0 8.1525 0 0
0 0 0 0 0 0 0 8.5626 0 13.0843
0 0 6.3313 0 0 6.5049 0 0 9.9939 0 0 0
0 7.4457 5.6057 0 8.1259 8.281 5.3428 6.5002 5.6739 0 8.4942 11.8075 0

R-vine dependence during the post-GFC period (2009-2011)

In this last part of our analysis we look at the dependencies during the post-GFC period. Figure-9 gives the first two trees of the fitted R-vine model. It is interesting to observe that Japan, Korea

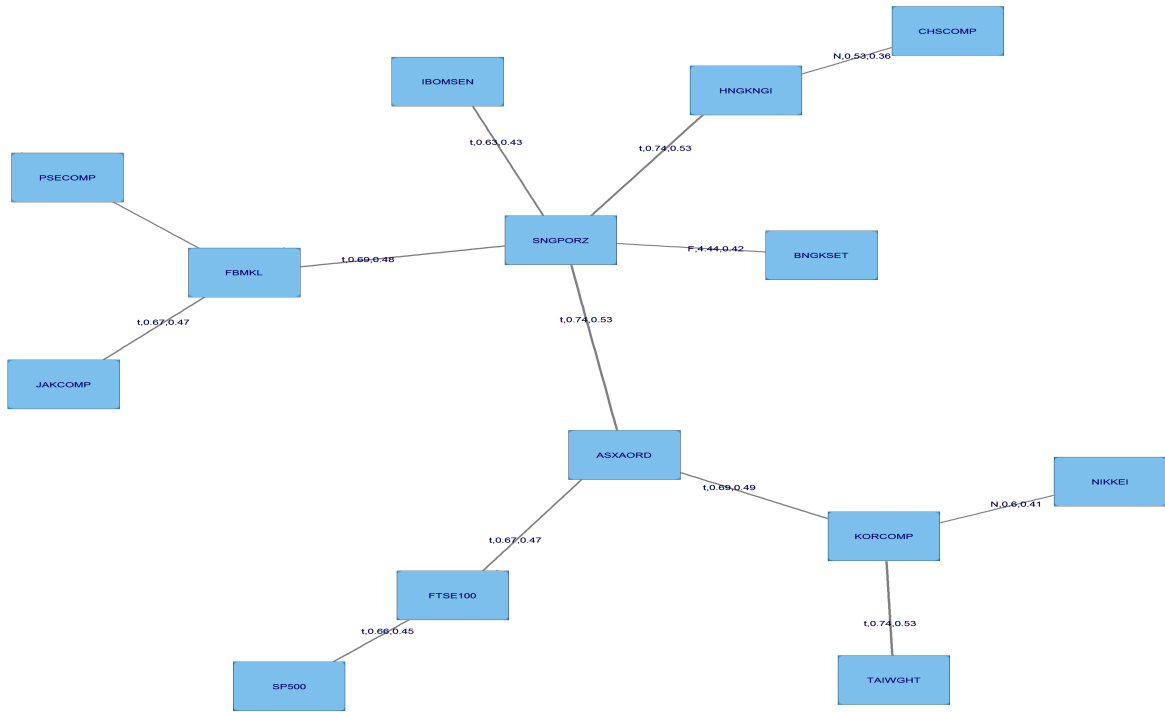
and Taiwan which showed more dependence with Hong Kong during the GFC period are now more closely related to Australia in the post-GFC period. This could be the outcome of the rather strong Australian economy as compared to the US and the UK during the period surrounding the GFC. The common feature from the pre-GFC, GFC and post-GFC period is the dependence of US-UK-Australia which remained intact during and after the GFC.

Table 18: R-Vine Copula Specification Matrix (post-GFC)

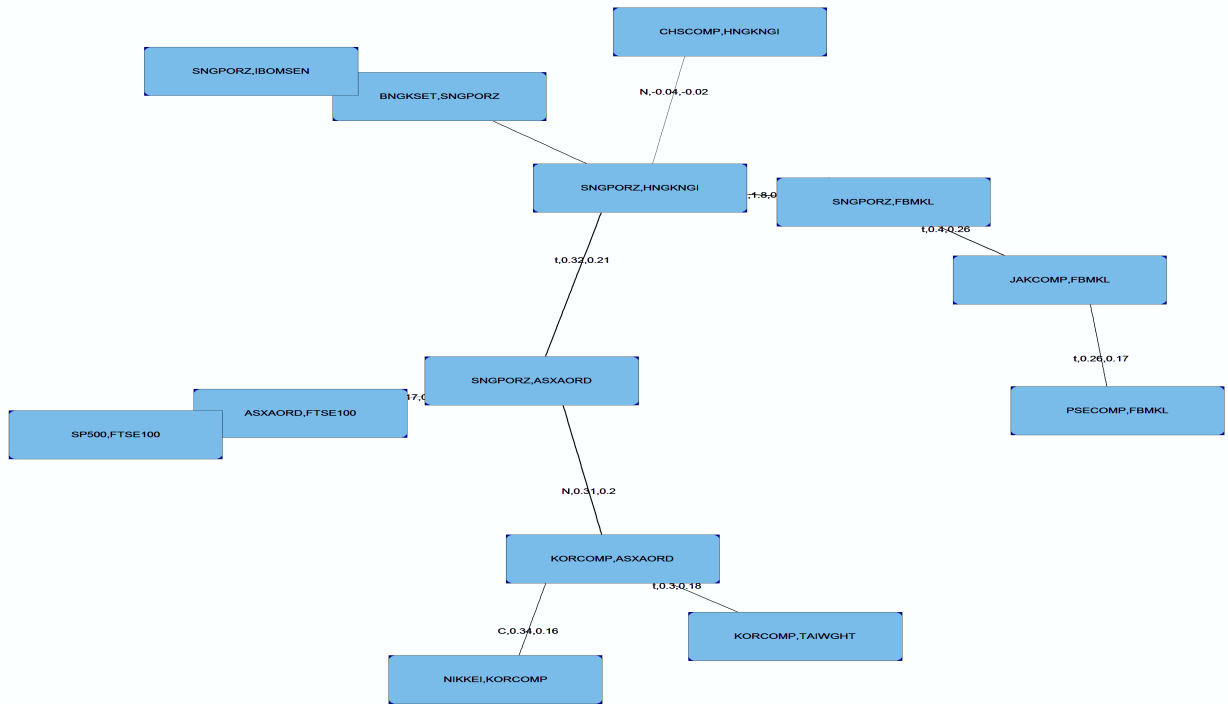
10													
7	3												
13	7	7											
14	13	5	13										
1	14	2	5	14									
9	1	11	2	5	1								
6	9	13	11	2	5	5							
12	6	14	8	11	2	9	2						
5	12	1	1	8	11	6	9	11					
2	5	9	9	1	8	12	6	9	8				
11	2	6	6	9	4	8	12	6	9	9			
8	11	12	4	6	9	4	8	12	6	4	4		
3	8	4	12	4	12	2	4	8	12	12	6	12	
4	4	8	14	12	6	11	11	4	4	6	12	6	6

Table 19: R-Vine Copula Type Matrix (post-GFC)

5													
5	5												
3	1	1											
5	1	3	5										
5	5	5	5	1									
1	5	5	5	1	3								
2	5	1	2	5	6	5							
5	1	5	5	1	4	3	2						
3	3	5	1	5	2	1	5	2					
3	5	4	2	2	5	3	5	2	2				
4	3	3	6	5	5	5	5	4	1	3			
4	4	1	5	2	3	2	2	5	2	2	1		
2	5	1	2	2	1	1	2	2	2	2	2	2	



(a) R-Vine (Tree-1)



(b) R-Vine (Tree-2)

Figure 9: R-Vine Tree 1 & 2 , post-GFC Period

The changes in other dependencies can be seen in table-18 with the selected copulas given in table-19. Table-19 shows that not only has the dependence structure changed after the GFC but so has the nature of the dependence among the financial markets. For example, the conditional dependence of 10,7 (India and China) in column-1 of table-14 is not just conditioned on a different set of markets but is also fitted with a Frank copula (5) which has negative dependence as given in table-20. Similar changes can be observed from these tables.

Table 20: R-Vine Copula Parameters (GFC)

(a) Copula Parameter-1														(b) Copula Parameter-2													
0.1101														0													
-0.4808	-0.0048													0	0												
-0.2363	0.0602	-0.051												0	0	0											
-0.0246	0.1709	-0.4535	-0.7188											0	0	0	0										
0.3847	0.0454	1.0496	-0.039	-0.8619										0	0	0	0	0									
0.1661	0.1034	0.0554	-0.2187	-1.7313	-0.1624									0	0	0	0	0	0								
0.1487	1.0004	0.0229	-0.2307	-0.0696	-0.0231	0.0529								8.3619	0	0	5.7144	0	0	0							
0.0639	0.14	0.1352	0.0831	0.6269	1.0784	-0.0469	-0.0667							0	0	0	0	0	0	0	0						
0.4159	-0.0198	0.91	0.0826	0.2274	0.1738	0.1375	1.0183	0.226						0	0	0	0	0	13.9309	0	0	9.3577					
0.0622	1.0262	0.1669	-0.3775	1.0351	0.1373	0.1531	0.103	-0.4164	-0.1363					0	0	0	10.5918	15.6741	0	0	0	14.2151	7.8627				
0.5962	0.2197	0.1749	-0.089	0.1216	1.111	1.9515	1.5102	0.202	-0.6144	0.0393				0	0	0	0	0	0	0	0	0	0	0			
1.1892	0.2202	0.2252	0.446	0.3281	0.2908	0.2876	1.773	0.2327	1.097	1.806	-0.675			0	0	0	14.3821	0	6.4536	8.6405	0	16.9942	10.8335	0			
0.6495	0.6471	0.655	0.5037	0.5668	0.6946	0.6599	0.7021	0.7925	0.555	0.7016	0.5916	0.561		16.4738	0	0	6.7051	11.3396	0	0	6.0709	10.0393	9.053	13.2406	11.5918	13.6918	

5 Conclusion

In this paper we put to work the recently developed R-vine copulas to analyse the dependence structure among 14 international stock markets. We analysed the co-dependencies of the financial markets of 11 Asian and 3 other developed countries during two different financial crisis events of the AFC and the GFC which happened a decade apart. The results suggest that the dependencies change in a complex manner and there is evidence of increase in tail dependence during the period of financial crisis, which is consistent with the existence of the larger tails to the distributions of returns usually found during times of financial distress. One of the attractions of this approach to risk modelling is the flexibility available in the choice of distributions used to model co-dependencies. The results also showed the change in the dependence structure of the developed markets of Australia, US and UK with the other Asian markets.

Looking at the empirical results from the period around the AFC and the GFC, a schematic shift can be seen in the dependence of the markets of the US and the UK with other Asian markets. The markets of Australia, the US and the UK had a stronger dependence with the Asian markets of Malaysia and Hong Kong in the pre-AFC period which changed slightly during the peak of the AFC and these three markets became more dependent on Hong Kong. This remained the case even

in the post-AFC period but it had completely changed by the pre-GFC period. Starting from the pre-GFC period Australia-US-UK shared strong dependence with Singapore which remained the case for the GFC and post-GFC periods. This is one of the many important conclusions which can be drawn from the flexible R-vine modelling as illustrated in this study.

The study can be furthered with a risk modelling application of the calculated dependencies. The static nature of this study, which is also one of the major limitation of this study can also be improved using vine-copula in dynamic modelling of risk and dependencies.

References

- Aas, K., & Berg, D. (2009). Models for construction of multivariate dependence - a comparison study. *The European Journal of Finance*, 15(7-8), 639-659.
- Aas, K., Czado, C., Frigessi, A., & Bakken, H. (2009). Pair-copula constructions of multiple dependence. *Insurance: Mathematics and Economics*, 44(2), 182 - 198.
- Alexandar, C. (Ed.). (2008). *Market risk analysis: Practical financial econometrics* (Vol. II). Wiley Publishing.
- Allen, D. E., Ashraf, M. A., McAleer, M., Powell, R. J., & Singh, A. K. (2013). Financial dependence analysis: applications of vine copulas. *Statistica Neerlandica*, 67(4), 403–435.
- Ang, A., & Chen, J. (2002). Asymmetric correlations of equity portfolios. *Journal of Financial Economics*, 63(3), 443 - 494.
- Bedford, T., & Cooke, R. M. (2001). Probability density decomposition for conditionally dependent random variables modeled by vines. *Annals of Mathematics and Artificial intelligence*, 32(1-4), 245–268.
- Bedford, T., & Cooke, R. M. (2002). Vines—a new graphical model for dependent random variables. *The Annals of Statistics*, 30(4), 1031–1068.
- Brechmann, E. C. (2010). *Truncated and simplified regular vines and their applications*. Diploma.
- Brechmann, E. C., & Czado, C. (2013). Risk management with high-dimensional vine copulas: An analysis of the euro stxx 50. *Statistics & Risk Modeling*, 4(30), 307-342.
- Brechmann, E. C., Czado, C., & Aas, K. (2012). Truncated regular vines in high dimensions with application to financial data. *Canadian Journal of Statistics*, 40(1), 68–85.
- Brechmann, E. C., Czado, C., & Paterlini, S. (2014). Flexible dependence modeling of operational risk losses and its impact on total capital requirements. *Journal of Banking & Finance*, 40(0), 271 - 285.
- Cheung, W. (2009). *Copula: A primer for fund managers*. Retrieved from <http://ssrn.com/abstract=1456980>

- Chow, G. C., Huang, S., & Niu, L. (2011). *Econometric analysis of stock price co-movement in the economic integration of east asia*. (Working Paper)
- Cooke, R. M., Joe, H., Aas, K., Kurowicka, D., & Joe, H. (2011). Vines arise. In D. Kurowicka & H. Joe (Eds.), *Dependence modeling: Vine copula handbook*.
- Dissmann, J. (2010). Statistical inference for regular vines and application. *Diploma thesis, Technische Universität München*.
- Dissmann, J., Brechmann, E. C., Czado, C., & Kurowicka, D. (2013, March). Selecting and estimating regular vine copulae and application to financial returns. *Comput. Stat. Data Anal.*, 59, 52–69.
- Embrechts, P., McNeil, A., & Straumann, D. (2002). Correlation and dependence in risk management: properties and pitfalls. In M. Dempster (Ed.), *Risk management: Value at risk and beyond* (p. 176-223). Cambridge University Press.
- Fischer, M., Kock, C., Schluter, S., & Weigert, F. (2009). An empirical analysis of multivariate copula models. *Quantitative Finance*, 9(7), 839-854.
- Ghosh, A., Saidi, R., & Johnson, K. H. (1999). Who moves the asia-pacific stock markets-us or japan? empirical evidence based on the theory of cointegration. *Financial review*, 34(1), 159–169.
- Heinen, A., & Valdesogo, A., Robles. (2009). *Asymmetric capm dependence for large dimensions: the canonical vine autoregressive model* (CORE Discussion Papers No. 2009069). Université catholique de Louvain, Center for Operations Research and Econometrics (CORE).
- Joe, H. (1996). Families of m-variate distributions with given margins and m(m-1)/2 bivariate dependence parameters. *Lecture Notes-Monograph Series*, 28, 120–141.
- Joe, H. (Ed.). (1997). *Multivariate models and dependence concepts*. Chapman and Hall.
- Johnson, R., & Soenen, L. (2002). Asian economic integration and stock market comovement. *Journal of Financial Research*, 25(1), 141–157.
- Karmann, A., & Herrera, R. (2014). Volatility contagion in the asian crisis: New evidence of volatility tail dependence. *Review of Development Economics*, 18(2), 354–371.
- Kurowicka, D. (2011). Optimal truncation of vines. In D. Kurowicka & H. Joe (Eds.), *Dependence modeling: Vine copula handbook*.
- Kurowicka, D., & Cooke, R. (2003). A parameterization of positive definite matrices in terms of partial correlation vines. *Linear Algebra and its Applications*, 372, 225–251.
- Kurowicka, D., & Cooke, R. M. (2006). *Uncertainty analysis with high dimensional dependence modelling*. John Wiley & Sons.
- Kurowicka, D., & Joe, H. (2011). *Dependence modeling: Vine copula handbook*. World Scientific.
- Longin, F., & Solnik, B. (2001). Extreme correlation of international equity markets. *The Journal of Finance*, 56(2), 649–676.
- Low, R. K. Y., Alcock, J., Faff, R., & Brailsford, T. (2013). Canonical vine copulas in the context of modern portfolio management: Are they worth it? *Journal of Banking & Finance*, 37(8),

3085 - 3099.

- Mandelbrot, B. (1963). The variation of certain speculative prices. *The Journal of Business*, 36, 394-419.
- Masih, A. M., & Masih, R. (1999). Are asian stock market fluctuations due mainly to intra-regional contagion effects? evidence based on asian emerging stock markets. *Pacific-Basin Finance Journal*, 7(3-4), 251 - 282.
- Masih, R., & Masih, A. M. (2001). Long and short term dynamic causal transmission amongst international stock markets. *Journal of international Money and Finance*, 20(4), 563-587.
- Nelsen, R. B. (1999). *Introduction to copulas*. Springer Verlag.
- Ng, A. (2000). Volatility spillover effects from japan and the us to the pacific basin. *Journal of international money and finance*, 19(2), 207-233.
- Palamalai, S., M, K., & Devakumar, C. (2013). Stock market linkages in emerging asia-pacific markets. *SAGE Open*, 3(4).
- Park, J. W. (2010). Comovement of asian stock markets and the u.s. influence. *Global Economy and Finance Journal*, 3(2), 76-88.
- Patton, A. J. (2006). Modelling asymmetric exchange rate dependence. *International Economic Review*, 47(2), 527-556.
- Schirmacher, D., & Schirmacher, E. (2008). *Multivariate dependence modeling using pair-copulas* (Tech. Rep.). Enterprise Risk Management Symposium.
- Sklar, A. (1959). Fonctions de repartition a n dimensions et leurs marges. *Publications de l'Institut de Statistique de L'Universite de Paris*, 8, 229-231.
- Sriboonchitta, S., Liu, J., Kreinovich, V., & Nguyen, H. (2014). A vine copula approach for analyzing financial risk and co-movement of the indonesian, philippine and thailand stock markets. In V.-N. Huynh, V. Kreinovich, & S. Sriboonchitta (Eds.), *Modeling dependence in econometrics* (Vol. 251, p. 245-257). Springer International Publishing.